NXGEN Technology AG

Technical Specification: Integration of Hikvision NVRs & IP Cameras with Genesis

Table of contents

- **Overview**
- Integration Options
  - **ISAPI (HTTP/HTTPS REST API) – Primary Integration**
  - **RTSP (Real-Time Streaming Protocol) – Live/Playback Video**
- Connection Model
  - **Video Stream - On-demand connection**
  - Alarms - ISAPI Based Alarm Subscription (Persistent connection)
    - **Connection type**
    - **Behavior**
    - Alarm Data Model
    - **Advantages**
  - Alarms - Webhook/Alarm Server Based Alarm reception (On-demand connection)
    - **Supported Devices**
    - **Behavior**
    - **Alarm Data Model**
    - **Advantages**
    - How to Check if Alarm Server (Webhook) Support Is Available
  - Alarms - Active Polling (Periodic Connection)
    - When Polling Is Required
    - **Polling Frequency and Platform Behavior**
    - Limitations of Active Polling

Summary: ISAPI (AlertStream) vs Webhooks (Alarm Server) vs Active Polling

Overview

Genesis integrates with Hikvision video devices including IP cameras, NVRs, and DVRs using industry-standard protocols. The integration supports live video, playback, metadata reception, alarm ingestion, and device management. Depending on system requirements, multiple integration methods can be used simultaneously.

Integration Options

Hikvision devices support several integration mechanisms. The platform uses the following:

ISAPI (HTTP/HTTPS REST API) – Primary Integration

- Protocol: Proprietary REST-like interface
- Transport: HTTP(s)
- Used for:
  - Alarm subscription (event push)
  - Event notifications (JSON/XML + binary JPEG)
  - Get device configuration
  - PTZ commands
  - Output Relay Trigger
  - Alarm Snapshot retrieval

RTSP (Real-Time Streaming Protocol) – Live/Playback Video

- Protocol: Standards-based streaming protocol
- Transport: TCP/UDP over RTP
- Used for Live/Playback video streaming from Hikvision devices
- Platform connects to:
  - `rtsp://<ip>/Streaming/Channels/101` (main stream)

- `rtsp://<ip>/Streaming/Channels/102` (substream)
- `rtsp://<ip>/Streaming/tracks/101/starttime=<<startTime>>&endtime=<<endTime>>` (playback stream)

Connection Model

Video Stream - On-demand connection

Live video uses **on-demand RTSP connection**:

- Connection opened only when the operator views a camera
- Connection opened when Genesis needs to pull alarm images based on the alarm reporting time.
- Closed when not in use.
- No permanent video connection.

Purpose: Efficient use of bandwidth and device resources.

Alarms - ISAPI Based Alarm Subscription (Persistent connection)

Alarms use **persistent HTTP connection** via ISAPI:

1. /ISAPI/Event/notification/alertStream

Connection type

Long-lived TCP connection (HTTP keep-alive), where the device streams events continuously.

Behavior

- Platform initiates a persistent connection to the Hikvision device.
- Device sends events in a mixed-content stream:
  - XML/JSON metadata

    - Optional JPEG snapshot appended (JFIF format)
- When connection drops, platform automatically re-subscribes

Alarm Data Model

Each event packet contains:

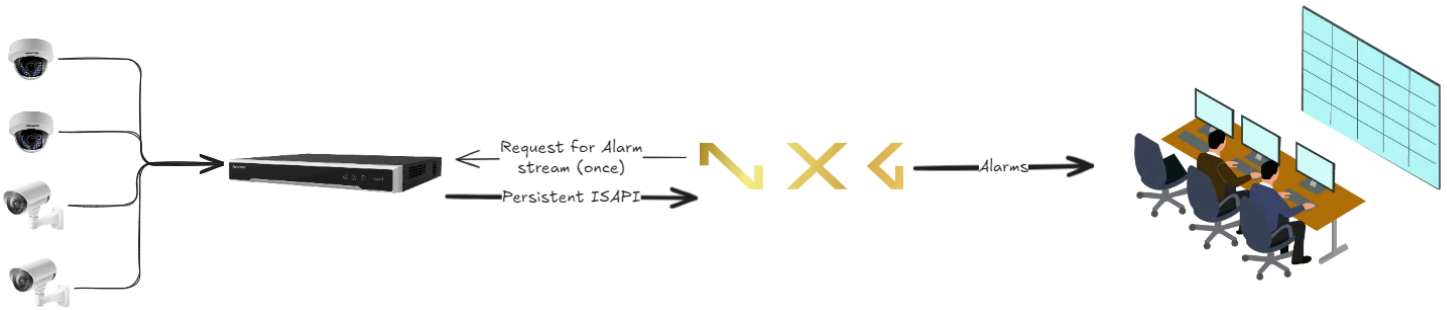
- XML metadata block (eventType, channelID, timeStamp)
- Optional binary JPEG snapshot (JFIF)
- Boundary-less mixed content (not multipart)
- Requires custom parsing (we extract metadata + JPEG buffer)

Examples of alarm types:

- Motion detection
- Line crossing / Intrusion
- Smart events (region entrance, trespass)
- Video loss
- Tampering
- VCA metadata
- Face detection, etc..

Advantages

- Real-time (<500 ms)
- No polling required
- Supports image attachment



Alarms - Webhook/Alarm Server Based Alarm reception (On-demand connection)

The Alarm Server feature allows supported Hikvision IP cameras to actively push alarm data to a platform-defined URL using HTTP/HTTPS. Unlike ISAPI alertStream, the connection is **initiated by the camera**, not the platform. Some Hikvision **IP cameras** (model-dependent) support a **Webhook-style Alarm Server**, where the camera **actively pushes alarm events** to a URL defined by the integrator. This operates differently from the traditional ISAPI `alertStream` subscription and is useful for cloud-based systems where the camera can initiate outbound connections.

Supported Devices

- Many **Hikvision IP cameras** running recent firmware (AcuSense, DeepInView, ColorVu)
- **Typically NOT supported** on:
  - DVRs
  - NVRs
  - Older series of cameras

Behavior

- Genesis Platform exposes a public HTTP/HTTPS endpoint.
- Camera sends an **HTTP POST request** to the configured URL whenever an event occurs.
- Each POST contains:
  - JSON or XML metadata describing the event
  - Optional snapshot image (inline base64 or multipart JPEG depending on model)
- The POST request is **one-shot per event**, not a persistent connection.
- Platform acknowledges with HTTP 200 OK.
- If delivery fails (network unreachable), the camera retries based on firmware logic (typically exponential retry).

Alarm Data Model

Each webhook POST contains structured metadata describing the event. Depending on camera model and firmware, the POST body may be:

- **JSON**
- **XML**
- **multipart/form-data** (snapshot + metadata)
- **JSON with embedded Base64 image**

Each webhook event typically includes:

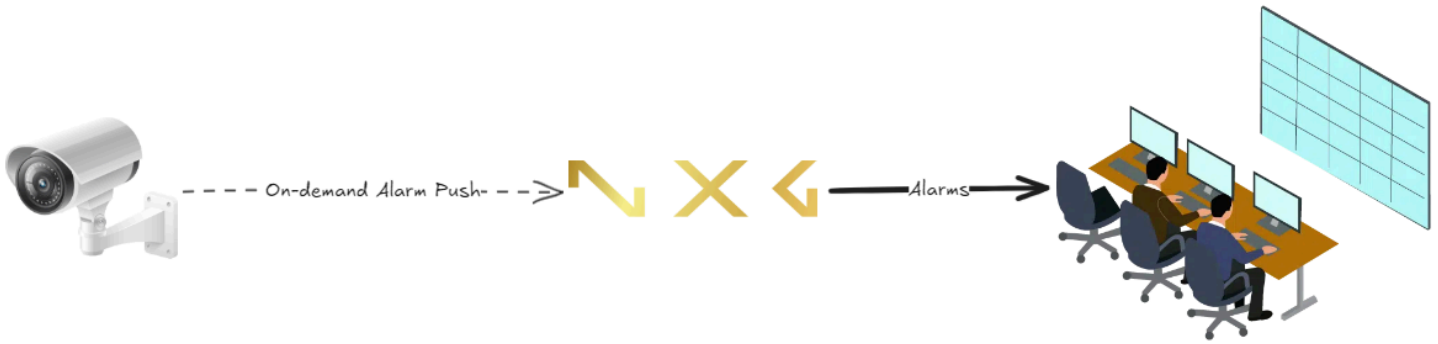
- `eventType`
- `channelID`
- `eventState` (active/inactive)
- `eventTime` (device local time)
- `deviceSN` (serial number)
- `channelName`
- Optional:
    - Snapshot image (Base64 or binary JPEG)
    - Object metadata (human/vehicle classification)
    - Target bounding boxes (on some AI models)
    - Rule Name (line crossing, region intrusion)

Examples of alarm types supported by Webhook:

- Motion detection
- Line crossing
- Intrusion detection
- Region entrance / exit
- Human / vehicle detection
- Face detection
- Tampering detection

Advantages

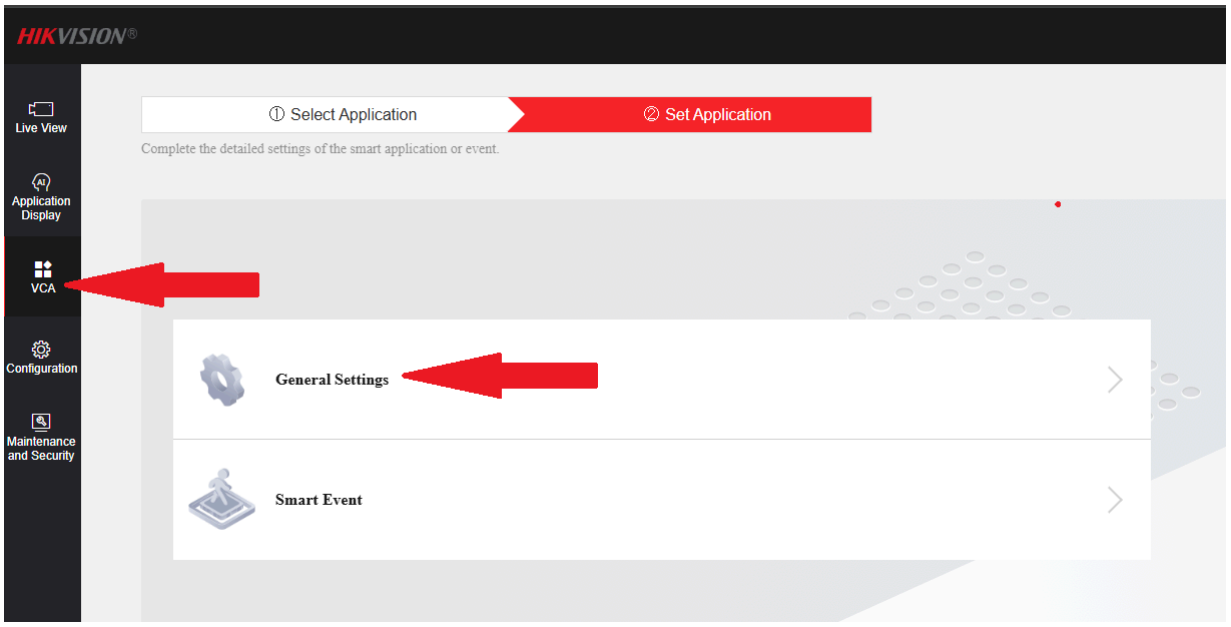
- **Camera-initiated delivery**
  - No need for persistent or on-demand Genesis platform to device connection.
  - Suitable for cloud platforms (**no port forwarding needed**).
- **Real-time performance**
  - Typically <500 ms latency
- **Supports snapshot delivery**
  - Base64 or binary JPEG depending on model
- **Firewall/NAT friendly**
  - Outbound-only connection from camera
- **Flexible payload formats**
  - JSON / XML / multipart

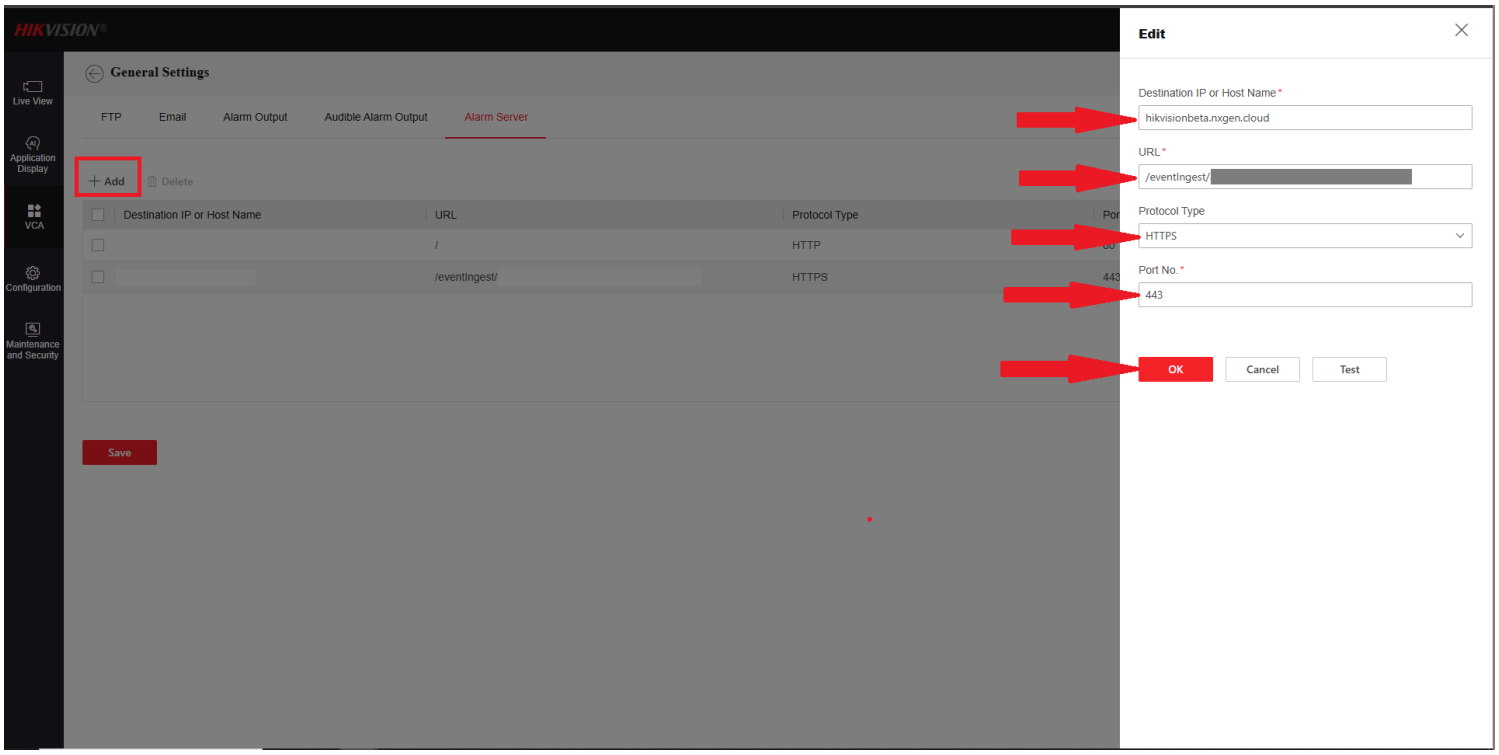


How to Check if Alarm Server (Webhook) Support Is Available

- Login to Camera's Web UI
- Navigate to VCA (or equivalent smart alert)-> General Settings -> Alarm Server
- Click "+Add"
- The popup should show the protocol type as HTTPS/HTTP.
- If you are seeing the above protocol type, then the IP Camera supports Alarm Server option

Screenshots below:





Alarms - Active Polling (Periodic Connection)

Some older or OEM-manufactured Hikvision devices do not support real-time alarm push mechanisms such as **ISAPI alertStream** or **Alarm Server (webhook)**. For these devices, the platform uses **active polling** as a fallback method to retrieve event information.

Active polling is used **only when neither ISAPI alarm streaming nor webhooks are available**, due to its limitations.

When Polling Is Required

Older Hikvision Firmware

Models manufactured before ISAPI became standard may not support:

- `/ISAPI/Event/notification/alertStream`
- `/ISAPI/Event/notification/httpHosts`
- `/ISAPI/System/capabilities` extended schema

These older devices expose only limited CGI-based endpoints or SOAP-style RPC interfaces.

Hikvision OEM Brands (Feature-Restricted Models)

Several OEMs rebrand Hikvision hardware but **disable Hikvision proprietary APIs** such as:

- ISAPI
- Alarm Server
- VCA metadata push

Examples include (but not limited to):

- **ABUS**
- **LTS**
- **Truvision Hikvision OEM**
- **TrendNet Hikvision OEM**
- **HiLook (some low-end models)**

These OEM models often:

- Block `/ISAPI` endpoints
- Provide only basic ONVIF functionality
- Do NOT send metadata or snapshots
- Do NOT support event push

As a result, the platform must resort to polling basic status endpoints.

Budget / Low-Energy / Entry-Level Models

Certain ultra-low-cost cameras disable advanced APIs to save firmware space.
Characteristics:

- ONVIF-only

- No event push
- Limited VCA
- No alertStream

Polling is the only viable integration method.

Polling Frequency and Platform Behavior

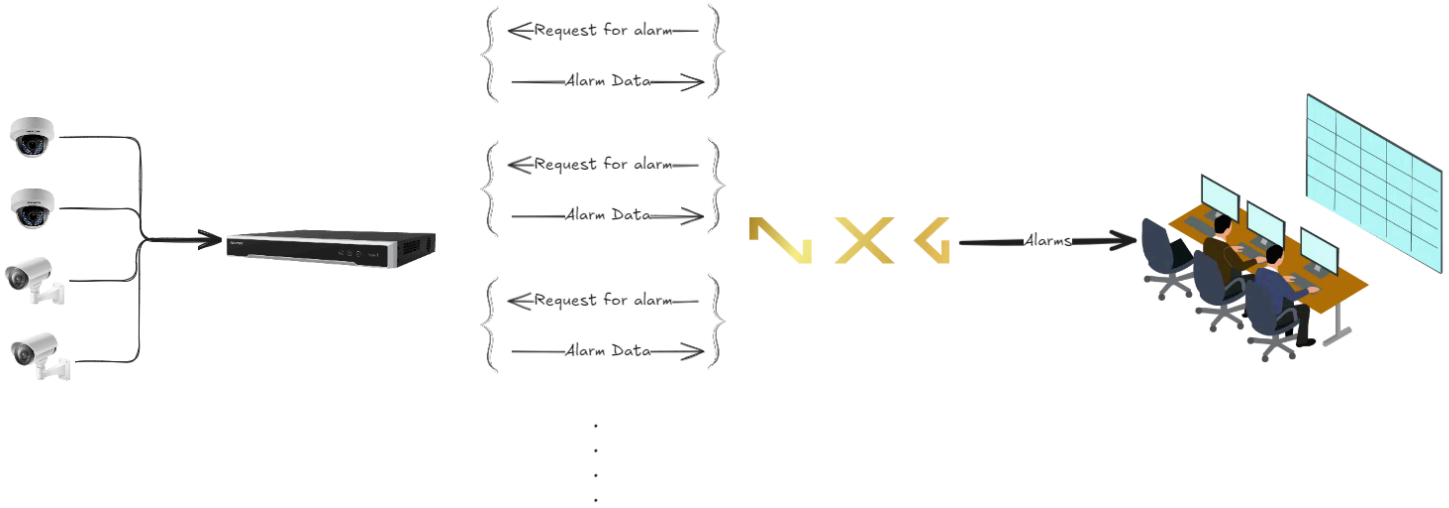
Polling runs at an interval of once in a minute.
Platform logic includes:

- Map OEM event names to normalized platform event types
- Fetch snapshot only on event trigger (when supported)
- Gracefully handle devices with unstable/missing endpoints

Polling is bandwidth- and CPU-intensive; therefore used **only when no push integration exists**.

Limitations of Active Polling

- Higher Latency - Events may be detected 1 to 1.5 minutes after they occur.
- Snapshot not guaranteed.
- Limited event types.
- Inconsistent firmware



Summary: ISAPI (AlertStream) vs Webhooks (Alarm Server) vs Active Polling

Feature	ISAPI Alert Stream	Alarm Server (Web Hooks)	Active Polling

Connection Direction	Genesis to Device	Device to Genesis	Genesis to Device
Connection Type	Long lived persistent stream - PUSH model	Single POST API call per event (Non persistent) - PUSH model	Periodic GET API Call to device - Pull/Poll model
Payload	Mixed XML + Jpeg (not always)	JSON/XML + Jpeg (not always)	JSON/XML
Image Format	Single Binary JPEG	Single Base64 or Multipart JPEG	No snapshot
Device Support	All DVRs, NVRs and IP Cameras	Some IP Cameras only	Old + OEM models
NAT/Firewall	Need port-forwarding	Works without inbound ports	Need port-forwarding
Reliability	High	High	Low
Preferred	Highly preferred	Highly preferred	Not preferred
Latency	< 500 ms (alarm reception from device to genesis)	< 500 ms (alarm reception from device to genesis)	> 1 minute